

FORENSIC ANALYSIS REPORT: DISTRIBUTED DENIAL OF SERVICE ATTACK

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Organization:	BSI (Bundesamt für Sicherheit in der Informationstechnik)
Malware:	Mirai Botnet Variant
Threat Level:	MEDIUM
File Hash:	1679091c5a880faf6fb5e6087eb1b2dc
Location:	Berlin

EXECUTIVE SUMMARY:

On 20th January 2025 between 09:15 and 11:47 CET, the web infrastructure of a major e-commerce platform based in Berlin experienced a significant Distributed Denial of Service (DDoS) attack. The attack originated from a Mirai botnet variant consisting of approximately 15,000 compromised IoT devices including IP cameras, DVRs, and home routers. Peak traffic reached 47 Gbps with 8.2 million packets per second, targeting both HTTP/HTTPS layers (L7) and network layer (L3/L4).

Analysis of packet captures revealed attack traffic originating from 67 different countries, with highest concentration from IP ranges in Brazil (AS28573), Vietnam (AS45899), and India (AS9829). The botnet employed reflection/amplification techniques using DNS, NTP, and SSDP protocols. Malformed HTTP requests with randomized User-Agent strings were designed to bypass basic rate-limiting mechanisms. The attack signature matches known Mirai variant "Satori" based on specific TCP flag combinations and payload characteristics.

Network forensics identified the C2 server coordinating the attack at IP address 162.244.83.47 (AS53667) communicating on non-standard port 7547. The attack was successfully mitigated through deployment of upstream scrubbing services and implementation of BPF filters to drop malicious traffic patterns. Post-incident analysis recommends enhanced monitoring of IoT device segments and implementation of micro-segmentation to prevent future botnet recruitment within the network perimeter.